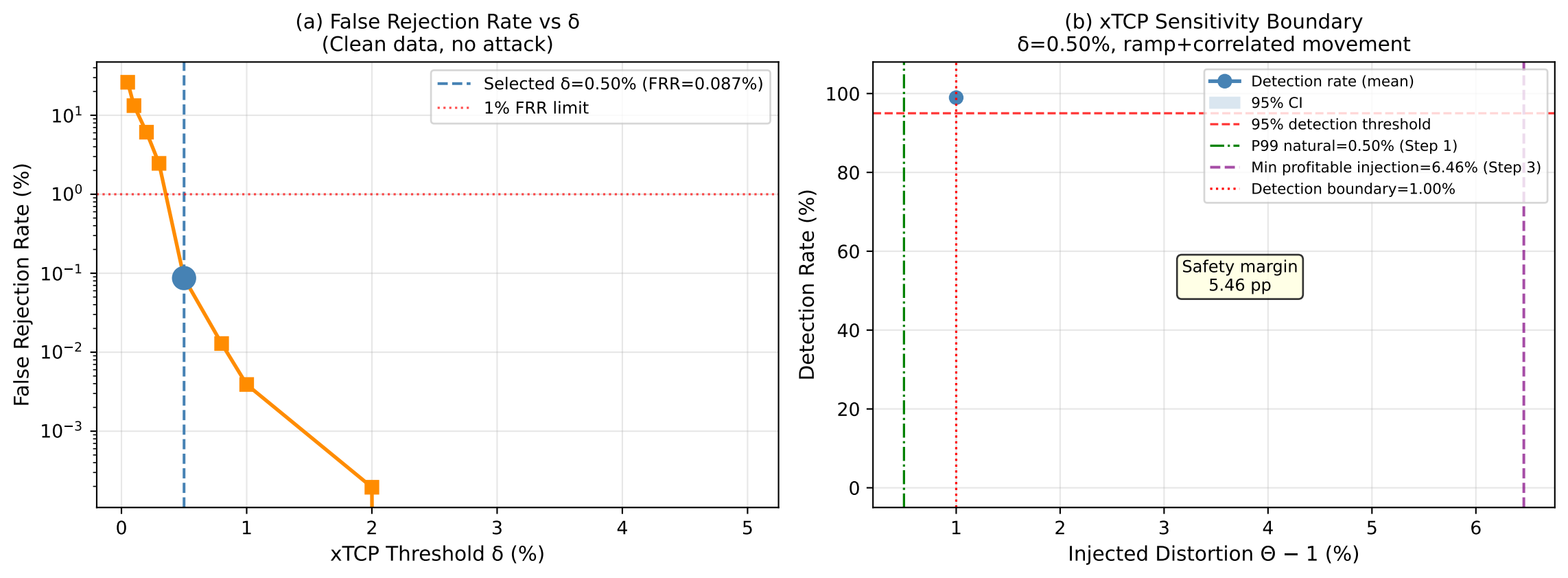
(a) False Rejection Rate vs δ
(Clean data, no attack)
Selected δ=0.50% (FRR=0.087%)
1% FRR limit
False Rejection Rate (%)
xTCP Threshold δ (%)
(b) xTCP Sensitivity Boundary
δ=0.50%, ramp+correlated movement
Detection rate (mean)
95% CI
95% detection threshold
P99 natural=0.50% (Step 1)
Min profitable injection=6.46% (Step 3)
Detection boundary=1.00%
Safety margin
5.46 pp
Detection Rate (%)
Injected Distortion Θ − 1 (%)